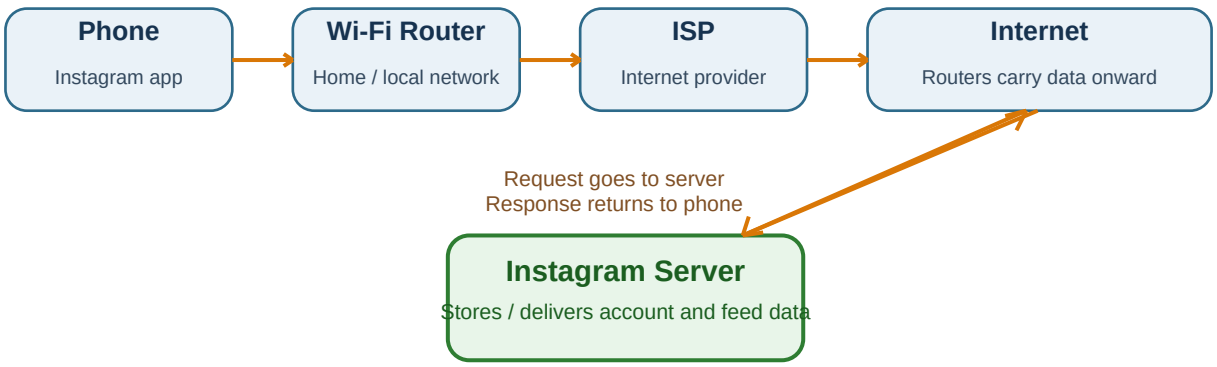
COMPUTER NETWORKING ASSIGNMENT

Internet, Network Types, Intranet, and App Data Flow

Student Name: _____
Class / Semester: _____
Subject: Computer Networking
Date: _____

Question 1: Diagram showing how a phone connects to Instagram

When a person opens Instagram on a phone, the phone sends requests through a local network and the Internet to Instagram's servers. The server processes the request and sends information such as posts, images, comments, or search results back to the phone.



Simplified path: actual Internet traffic may pass through several routers and data centres.

Labels and connections: The phone connects wirelessly to the Wi-Fi router. The router forwards traffic through the Internet Service Provider (ISP). The ISP connects the home network to the wider Internet. Internet routers carry the request to Instagram's server. The server sends a response back along the network path to the phone.

Question 1 — How the connection works

Step 1 — Phone and Instagram app

The user opens Instagram on a smartphone. The app prepares a request, for example, to load the home feed or search for an account. The phone uses Wi-Fi or mobile data to send the request.

Step 2 — Wi-Fi router (local network)

When Wi-Fi is used, the phone sends data wirelessly to the Wi-Fi router. The router connects devices in the home or college network and forwards traffic towards the Internet. The phone-to-router part is usually a wireless local area network (WLAN), which is a type of LAN.

Step 3 — ISP (Internet Service Provider)

The router passes the data to the ISP through the broadband connection. The ISP is the company that provides Internet access. It routes the data onward and helps connect the user's network to online services.

Step 4 — Internet routers and Instagram server

The request travels across the Internet through one or more routers and network links until it reaches Instagram’s systems. Instagram’s servers process the request and determine what information should be returned. The actual route can vary depending on network conditions and the service’s infrastructure.

Step 5 — Response returns to the phone

Instagram sends the requested data back through the Internet and the ISP, then through the Wi-Fi router to the phone. The app displays the result on the screen. Much of the traffic is protected using HTTPS encryption while travelling between the app and the service.

Important note: If the phone uses mobile data instead of Wi-Fi, the phone connects to a mobile network tower and the mobile operator’s network instead of the home Wi-Fi router. The request still reaches Instagram through the Internet.

Question 2: Five everyday devices that are part of a network

A network is a group of connected devices that can exchange data or share resources. The table below gives five common examples and the network types they commonly use.

Device	Network type	How it connects / example
Smartphone	WLAN / LAN, WAN, PAN	Uses Wi-Fi to join a home or college WLAN, or mobile data to access a WAN. It may also use Bluetooth PAN to connect to earbuds.
Laptop	LAN / WLAN; WAN via Internet	Connects to a router through Wi-Fi or Ethernet. It can access websites, online classes, shared printers, and files.
Smart TV	LAN / WLAN; WAN via Internet	Uses Wi-Fi or Ethernet to connect to the home network and stream online videos from services.
Wi-Fi router	LAN / WLAN; WAN uplink	Connects local devices together and links the local network to the ISP. It is a network device rather than an end-user device.
Smartwatch	PAN; sometimes WLAN / WAN	Often connects to a phone using Bluetooth PAN. Some models also connect directly to Wi-Fi or cellular networks.

Understanding the network terms

LAN (Local Area Network): Connects devices in a limited area such as a room, home, lab, or office. Ethernet is a common wired LAN connection.

WLAN (Wireless Local Area Network): A LAN that uses Wi-Fi instead of cables for the local connection.

WAN (Wide Area Network): Covers a large geographical area and connects networks over long distances. The Internet is the best-known example of a global network of networks.

PAN (Personal Area Network): Connects personal devices over a short range, often using Bluetooth—for example, a phone and wireless earbuds.

Note: A single device can use more than one network type. For example, a smartphone can use WLAN for Wi-Fi, WAN for mobile Internet access, and PAN for Bluetooth accessories.

Question 3: Difference between the Internet and an intranet

The **Internet** is a worldwide network that connects millions of private, public, educational, business, and government networks. It allows people to visit public websites, send emails, use social media, and access online services from different locations.

An **intranet** is a private network used within an organisation, such as a college, company, or hospital. It is intended for authorised users and may provide internal notices, staff/student resources, timetables, forms, or shared files. Access may require a college account, login, or connection to the college network.

Example from a college

Imagine a college has a student portal that shows internal notices, attendance, lab schedules, and course materials. If the portal is intended only for students and staff with authorised logins, it is an example of an intranet service. When a student opens a public website such as Instagram or Wikipedia, the student is using the Internet.

Feature	Internet	Intranet
Access	Generally public, although individual services may require accounts.	Restricted to authorised members of an organisation.
Purpose	Public communication and online services worldwide.	Internal information and resource sharing.
Example	Instagram, public news sites, online shopping.	College student portal or internal staff website.
Security	Uses many different security controls; users should still protect accounts and data.	Usually protected by organisation-managed logins and access rules.

Important difference: An intranet can use Internet technologies such as web browsers and network protocols, but it is private to an organisation. Some intranet services may be reachable from outside the college through a secure login or VPN; this does not make the intranet itself the public Internet.

Question 4: How data travels when searching in WhatsApp

Example: A user opens WhatsApp and searches for a contact or a message. The exact steps depend on what is being searched. Searching messages already stored on the phone may happen locally without sending the search to WhatsApp's servers. The following steps describe a typical online request, such as loading account or contact-related information.

Step-by-step data flow

Step 1 — Enter the search: The user types a name or keyword in the WhatsApp app and taps Search. The app interprets the action and prepares the information needed for the request.

Step 2 — Connect to a network: The phone sends data over Wi-Fi to a router, or over mobile data to a mobile network. The local network and the ISP/mobile operator provide a path to the Internet.

Step 3 — Reach the service: If the action requires an online request, data travels through Internet routers to the relevant WhatsApp service. The connection is protected with encryption in transit.

Step 4 — Process the request: The service processes the request and sends back the appropriate response. For a search of locally stored chats, the phone's app may do the searching on the device instead.

Step 5 — Display the result: The response returns through the Internet and the phone's network connection. WhatsApp displays the matching result or updated information on the screen.

In short (4–5 lines)

When I enter a search in WhatsApp, the app processes the keyword and checks the relevant data. If an online request is needed, my phone sends it through Wi-Fi and the router, or through mobile data, to the Internet. The request reaches WhatsApp's service, which processes it and sends a response back. The response travels through the network to my phone, and the app displays the result. Some message searches are performed directly on the phone and may not need a server request.

Conclusion

Networks allow devices to communicate and share information. A phone can connect to a local Wi-Fi network, reach the Internet through an ISP, and communicate with an app's online servers. The Internet is public and worldwide, while an intranet is a private network for an organisation. Different devices may use LAN, WLAN, WAN, or PAN depending on how they connect.